SOURCE FACT · OBSERVABLE EVENT · OPERATIONAL DECISION

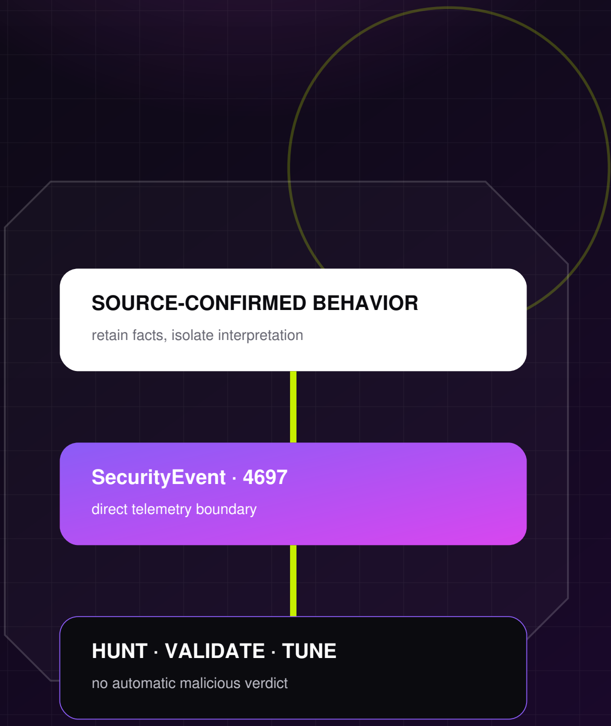
DETECTION THREAT LANDSCAPE

HONEYMYTE COOLCLIENT DRIVER SERVICE

Campaign hunt for the source-observed msagent kernel driver service and msagent.sys image.

HUNTING QUERY

CONFIDENCE: MEDIUM



REPORT DATE
15 August 2026

VERSION
v1.0

PRIMARY TELEMETRY
Microsoft Sentinel · SecurityEvent · 4697

ATT&CK
T1543.003

Evidence-led dossier. Query status remains an untested implementation sketch until executed in the target environment.

01

EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Hunt the exact driver service/image pair at creation time.

Kaspersky observed CoolClient extracting `msagent.sys`, replacing any existing `msagent` service, then creating and starting a new driver service. Windows Security Event 4697 directly records the installed service name and image path.

OPERATIONAL VALUE

High-value pre-rootkit evidence with explicit artifact limits.

The service/image pair is far more selective than a generic driver-install alert and appears before the driver can hide processes, files, Registry objects, modules and C2 addresses. Both strings are mutable, so the result remains a Medium-confidence campaign hunt.

9.0 — OPERATIONAL VALUE	01	
Medium — DETECTION CONFIDENCE	02	
Hunting — OPERATIONAL READINESS	03	

SOURCE-CONFIRMED FACTS

Observed behavior

- Published 14 August 2026; activity was observed in late 2025 and 2026.
- CoolClient writes msagent.sys and creates a driver service named msagent.
- The driver exposes IOCTL handlers for protection, hiding and C2-address filtering.

ANALYTICAL ASSESSMENT

What the evidence supports

- Event 4697 is direct service-installation evidence.
- The paired name and image basename reduce benign overlap.
- A match cannot establish driver load, rootkit operation or attribution.

DETECTION ASSUMPTIONS

What must be validated

- Audit Security System Extension is enabled and 4697 reaches Sentinel.
- SecurityEvent populates ServiceName and ServiceFileName.
- The source-observed service/image pair is not owned by approved software.

Confidence boundary

The report analyzes malware and observed intrusions, but does not provide tenant telemetry. The query is untested and cannot prove successful driver loading, IOCTL execution, process hiding, credential theft, C2 activity or HoneyMyte attribution.

02

THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01
Post-compromise deploy
PlugX deploys CoolClient components.

02
Privilege gate
CoolClient checks SCM access and SeTcbPrivilege.

03
Write driver
An embedded driver is written as msagent.sys.

04
Install service
The msagent driver service is replaced, created and started.

05
Enable stealth
IOCTLs register the process, C2 IP and protected paths.

Observable core
A Windows Security Event 4697 with ServiceName msagent and ServiceFileName ending in \msagent.sys.

Non-observable boundary
Driver contents, successful load/start, IOCTL traffic, hidden objects, network filtering, C2 outcome and actor attribution.

DETECTION HYPOTHESIS

CoolClient driver deployment should leave a new-service record.

If an attacker repeats the documented driver installation, SecurityEvent should show Event ID 4697 with ServiceName msagent and ServiceFileName ending in \msagent.sys. Legitimate activity can be separated through service ownership, complete path, signer, hash, certificate, installer identity and neighboring endpoint evidence.

REQUIRED TELEMETRY
SecurityEvent · 4697
- Windows Security Event 4697.
- Windows Security Events via AMA.
- ServiceName and ServiceFileName.

OPTIONAL ENRICHMENT
Context that improves disposition
- Service type/start type and subject identity.
- File hash, signer, certificate and origin.
- Driver-load, process, Registry and network telemetry.

UNAVAILABLE HERE
Do not infer
- IOCTL calls in native SecurityEvent.
- Proof of hidden processes/ files/Registry entries.
- Successful C2 or HoneyMyte attribution.

03

TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Sentinel · SecurityEvent · 4697

SecurityEvent documents EventID, ServiceName, ServiceFileName, service metadata, subject identity, host and EventRecordId. Microsoft documents Event 4697 as generated when a new service is installed and its image path is captured at creation time.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The event establishes service-installation metadata only. ServiceType is projected for triage but deliberately not filtered because the normalized Sentinel value must be validated locally.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Service event	`SecurityEvent / EventID`	4697 installation event.	REQUIRED
Service identity	`ServiceName`	Exact msagent service.	REQUIRED
Image path	`ServiceFileName`	Ends with msagent.sys.	REQUIRED
Host	`Computer, _ResourceId`	Asset pivot.	REQUIRED
Event identity	`TimeGenerated, EventRecordId`	Raw record pivot.	REQUIRED
Subject	`SubjectUserSid, SubjectUserName, SubjectDomainName`	Installer identity.	OPTIONAL
Logon context	`SubjectLogonId`	Neighboring activity pivot.	OPTIONAL
Service metadata	`ServiceType, ServiceStartType`	Triage; not filtered.	OPTIONAL

Data quality requirements

- Confirm Event 4697 collection on relevant Windows assets.
- Create a controlled driver service and inspect exact field values.
- Verify ServiceFileName retains the complete path and basename.
- Review historical collisions without a rarity threshold.

Known failure conditions

- Audit Security System Extension is disabled.
- The connector drops 4697 or required fields.
- The actor renames either artifact or modifies an existing service.
- The driver is loaded through another mechanism.

04

DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find the source-observed CoolClient driver service pair.

Select new-service events only when Event 4697 records both the msagent service name and an image path ending in msagent.sys.

ANALYTICAL LOGIC

Direct filters and investigable output.

Three direct predicates: Event ID, exact case-insensitive service name and image-path suffix. No hash, domain, threshold, join, sequence, aggregation, rarity, regex, external lookup or automatic attribution.

KQL · UNTESTED HUNT

```
// Hunting query - Untested implementation sketch
// Primary platform: Microsoft Sentinel / Log Analytics
// Required connector: Windows Security Events via AMA
// Required audit policy: Audit Security System Extension (Success)
// Evidence: Kaspersky observed CoolClient creating the msagent driver service
// and writing the driver as msagent.sys. A match is a campaign lead, not proof
// that the driver loaded, its IOCTLs ran, or HoneyMyte operated the host.
SecurityEvent
| where EventID == 4697
| where ServiceName =~ "msagent"
| where ServiceFileName endswith @"\msagent.sys"
| project
    TimeGenerated,
    Computer,
    EventRecordId,
    ServiceName,
    ServiceFileName,
    ServiceType,
    ServiceStartType,
    SubjectUserSid,
    SubjectUserName,
    SubjectDomainName,
    SubjectLogonId,
    _ResourceId
| order by TimeGenerated desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	Computer + EventRecordId + TimeGenerated	Manual triage

What a match means

The host recorded creation of the source-observed CoolClient driver service/image pair and requires immediate service, file and installer review.

What a match does not prove

Successful driver start/load, rootkit behavior, IOCTL use, credential theft, C2 traffic, persistence success or HoneyMyte attribution.

05

TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

01 **Confirm the raw event.**
Verify both strings, complete path, service metadata, host, subject and record ID.

02 **Validate ownership.**
Check driver inventory, software owner, package provenance and change record.

03 **Collect the driver.**
Acquire msagent.sys and calculate hashes, signer state and certificate details.

04 **Rebuild installation.**
Review file creation, SCM access, service start and nearby process execution.

05 **Scope rootkit effects.**
Compare EDR, memory, filesystem, Registry and network views for visibility gaps.

Escalation gate

Escalate when the pair is unowned or the driver certificate/hash is unknown, especially with CoolClient files, fake Defender paths, Sangfor sideloading, suspicious Defender exclusions, synchost.exe injection or unexplained network visibility gaps.

Primary benign overlap

A legitimate product or lab could reuse generic msagent naming. Tune only with verified publisher, driver hash/provenance, approved owner and stable path; never suppress the service name globally.

INVESTIGATION PIVOTS

Host
Owner, role, exposure and EDR coverage.

Service
Current ImagePath, start mode and Registry key.

Driver
Hash, signer, certificate and load state.

Installer
Subject logon and initiating process.

Registry
SYSTEM\RNG and service configuration.

Network
C2 IP visibility across endpoint and network sensors.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.

06

FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Legitimate driver package	A vendor may use msagent naming.	Require verified publisher, catalog, owner and stable path.
Security research replay	A lab may reproduce the published pair.	Require approved analyst, isolated asset and test window.
Internal test driver	QA tooling may create disposable drivers.	Validate package identity and change record.

Baseline plan

- Search the full available 4697 retention.
- Inspect every matching raw event and resolved driver path.
- Record signer, hash, owner and certificate for legitimate matches.
- Do not add a count or rarity threshold.

Validation procedure

- Enable Audit Security System Extension in an isolated lab.
- Create/remove a disposable driver service matching the pair.
- Run both split negative cases.
- Compare SecurityEvent with endpoint driver-load ground truth.

TEST CASES

CASE	EXPECTED RESULT
Create controlled msagent driver service for msagent.sys	Expected match with complete fields.
Create msagent service for another image	No match.
Create another service for msagent.sys	No match.
Modify an existing msagent service	Expected miss; documents 4697 boundary.

Hunting
Initial query

Candidate
Current status

Production
Measured precision

Optimized
Stable feedback

Success

The controlled pair matches with complete event identity, both negative cases do not, and historical collisions are absent or explainable.

Rejection

Either required field is missing or approved software frequently reproduces the complete pair without a durable discriminator.

Failure

A controlled matching driver-service creation produces no usable SecurityEvent record.

ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1543.003	Create or Modify System Process: Windows Service	Directly covers creation of the msagent driver service used to load CoolClient kernel functionality.

Detection coverage summary

Strong for the exact source-observed pair at creation time; weak for renamed artifacts, existing-service modification, alternative driver loading or missing 4697 coverage.

Residual risk

The actor can rename the service/driver or use another loader. Rootkit effects and successful C2 require endpoint, memory, Registry and network validation.

ANALYST CONCLUSION

A selective pre-rootkit campaign hunt with a clear promotion path.

Kaspersky directly documents the service/image pair and Microsoft documents Event 4697 plus the required Sentinel fields. This supports a Medium-confidence Hunting query. Artifact mutability and lack of target-environment testing prevent Production classification.

REFERENCES

1. Kaspersky GReAT, “APT group HoneyMyte upgrades CoolClient,” 14 August 2026; activity observed late 2025 and 2026. hxxps://securelist[.]com/honeymyte-coolclient-driver-rootkit/121028/
2. Microsoft Learn, “SecurityEvent table,” updated 27 July 2026; accessed 15 August 2026. hxxps://learn[.]microsoft[.]com/en-us/azure/azure-monitor/reference/tables/securityevent
3. Microsoft Learn, “4697(S): A service was installed in the system,” accessed 15 August 2026. hxxps://learn[.]microsoft[.]com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4697
4. MITRE ATT&CK, T1543.003. hxxps://attack[.]mitre[.]org/techniques/T1543/003/

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	15 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.